organism). Parasites of medical Importance come under the Kingdom called Protista and Animalia. Protista includes the microscopic single- celled eukaryotes known as protozoa. In contrast, helminths are microscopic, multicellular worms possessing well differentiated tissues and complex organs belonging to the kingdom Animalia. Classification of medically important parasites is given in Flowchart 8.2.

8.1.5 Life Cycle of Parasites

Direct life cycle

The life cycle of parasite that requires only single host to complete its development, is called direct life cycle. Example: *Entamoeba histolytica* requires only human host to complete its life cycle.

Indirect life cycle

The life cycle of parasite that requires two or more species of hosts to complete its development, the life cycle is called as indirect life cycle. Example: Malarial parasite (Plasmodium spp.) requires both human host and mosquito to complete its life cycle.

8.1.6 Transmission of Parasites

It depends upon on Source or reservoir of infection, and mode of transmission.

1. Sources of infection

A. Human: Human is the source or reservoir for a majority of parasitic infection. The condition in which the infection is transmitted from one infected human to another human is called **anthroponoses**.

B. Animals: Animals act as the source of infection in many parasitic diseases. The condition where infection is transmitted from animals to humans is called **zoonoses**.

2. Mode of transmission

A. Oral transmission: This is through ingestion of contaminated food, water, vegetables, soiled fingers or fomites contaminated by faeces that contain the infective stage of parasite. This mode of transmission is referred to as faecal-oral route. Example: Cysts of *Entamoeba histolytica*.

B. Skin transmission: This is another important route. The infective larvae of hookworm enter the skin of persons walking bare footed on contaminated soil.

C. Vector transmission: It could be a biological or a mechanical means. Many parasitic diseases are transmitted by insect bite. Example: sandfly is vector for *Leishmania*.

D. Direct transmission by person to person contact. Frequently, *Entamoeba, Giardia* and *Trichomonas* are transmitted by sexual contact among homosexuals.

E. Vertical transmission: It is the transmission from mother to fetus. Example: Toxoplasmosis.

So far, we have learnt about the general introduction and classification of parasites. Now, let us learn a few important human parasites in detail.

Introduction to Protozoa

General characteristics of protozoa:

1. They are microscopic unicellular eukaryotes.
2. The single cell has a relatively complex internal structure and it performs

TN_GOVT_XII_Micro_Biology_CH08.indd 154 27-02-2019 14:52:58